Sinoda Gebrehiwot Tesfay

Software Engineer : Backend, ML Infrastructure & LLM Training : 6 Years Experience

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Software Engineer with 6 years across backend systems, real-time ML infrastructure, and LLM training and evaluation. Core strengths in **Python, FastAPI, Go, and React/Next.js**: async API design, distributed systems, and production AI applications.

PROFESSIONAL EXPERIENCE

LLM Trainer, AI Data Annotation & Model Evaluation : Turing — Feb 2026 – Jul 2026 · Remote

- Produced high-precision, frame-level video annotations for LLM and multimodal model training, capturing actions, context, and temporal relationships across visual sequences.
- Evaluated model performance through targeted prompt design and response validation, feeding actionable findings back into the improvement cycle.

Back End Developer : Eskalate — Mar 2023 – Feb 2026 · On-site

- Led the modernization of backend systems from Flask and MongoDB to an async **FastAPI + SQLAlchemy** architecture, reducing API latency by 35%.
- Engineered real-time ML infrastructure on **FastAPI WebSockets**, streaming live training telemetry to browsers at 20 FPS with experiment tracking and on-demand model endpoints.
- Built and shipped the recommendation engine behind **Bazar**, an AI-powered marketplace, using TensorFlow, scikit-learn, and an automated Python + Selenium data pipeline.

React Frontend Developer : Kuraztech — Feb 2021 – Mar 2023 · On-site

- Reduced application load times by **40%** through lazy loading, code splitting, and React/Next.js performance work.
- Played a key role in the migration to **Next.js SSR/SSG**, strengthening SEO and contributing to a 25% increase in user engagement.

Full Stack Developer, Internal Tooling : Eskalate — Apr 2020 – Feb 2021 · Part-time

- Built a Slack-integrated automation platform used by **200+ employees** across four time zones, streamlining Google Sheets workflows, and a timezone-aware notification system on Redis and Celery that cut missed deadlines by 30%.
- Deployed highly available **Go microservices on Kubernetes**, contributing to 99.9% uptime while reducing deployment times by 60%.

PROJECTS

RustV OS Lab Bare-Metal RISC-V Operating System

A no-std Rust kernel on QEMU: Sv39 virtual memory, preemptive scheduling, syscalls, and a virtio-MMIO DMA driver.

Rust · RISC-V Assembly · QEMU · Sv39 VM · Virtio-MMIO

Hulu Code Code Executor Web Application

Interactive multi-language code execution platform supporting 7+ languages (Python, JavaScript, SQL), with real-time code sharing and cloud-based execution that cuts environment setup time.

Python · JavaScript · SQL · Cloud Execution · Real-Time

ImmuStore DB Distributed Database Engine Built from Scratch

Pure-Python distributed database: copy-on-write B+ trees, crash-safe MVCC, a Redis RESP server, and Raft replication.

Python · asyncio · B+ Trees · MVCC · Raft · Redis RESP

AI-RAD Healthcare Screening & Records Platform

Hospital COVID-19 screening and records platform: **1st place**, 2020 UNDP & MINT Innovation Challenge (580,000 ETB grant).

React · Node.js · Firebase · Firestore · RBAC

TECHNICAL SKILLS

Backend	Python, FastAPI, Flask, SQLAlchemy, Go, REST APIs, WebSockets, Redis, Celery, MongoDB, SQLite
Frontend	React, Next.js, TypeScript, JavaScript, HTML, CSS, component architecture
AI / ML	LLM training, model evaluation, data annotation, TensorFlow, scikit-learn, recommender systems
DevOps	Docker, Kubernetes, CI/CD, Git, microservices, distributed systems, monitoring
Engineering	System architecture, database design, OOP, algorithms, data structures, performance tuning

EDUCATION

BSc Software Engineering
Addis Ababa University (AAU)
Sep 2014 – Jul 2019 · Addis Ababa, Ethiopia

DS & Algorithms Program
A2SV : Africa to Silicon Valley
Apr 2019 – Sep 2020 · Addis Ababa, Ethiopia